QEncode Benchmark Suite Specification

QEncode Benchmark Suite

A Reproducible Framework for Quantum Algorithm Evaluation

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1. Purpose

This document defines the **official specification of the QEncode Benchmark Suite v1**.

The specification establishes standardized benchmark configurations used to evaluate quantum algorithms in quantum chemistry workloads.

The goal of the specification is to ensure:

- reproducibility
- consistent experimental configurations
- comparable benchmark results
- transparent algorithm evaluation

Only benchmark runs that strictly follow this specification can be classified as **Certified Benchmark Results**.

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2. Benchmark Scope

The QEncode Benchmark Suite focuses on **variational quantum eigensolver (VQE) workloads** for quantum chemistry problems.

The benchmark suite evaluates the performance of algorithm configurations across several dimensions:

- fermion-to-qubit encoding methods
- ansatz circuit families
- execution environments
- measurement configurations
- optimization strategies

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3. Benchmark Problems

The benchmark suite includes a curated set of molecular Hamiltonians.

Molecule	Basis	Active Space
H_2	STO-3G	(2,2)
BeH_2	STO-3G	(4,4)

Geometry

The default equilibrium geometry must be used.

Hamiltonian Construction

Hamiltonians must be constructed using the specified:

- basis set
- active space
- molecular geometry

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4. Fermion-to-Qubit Encodings

The following encodings are supported in Benchmark Suite v1:

Encoding	Description
Jordan-Wigner	Standard fermion-to-qubit mapping
Bravyi-Kitaev	Balanced locality and parity mapping
Parity	Parity-based mapping

Each benchmark run must specify the encoding used.

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5. Ansatz Families

Benchmark Suite v1 includes two ansatz circuit families.

Ansatz	Description
UCCSD	Unitary Coupled Cluster with Singles and Doubles
Hardware Efficient Ansatz	Parameterized hardware-native circuit

Ansatz Parameters

UCCSD configuration:

repetitions: 1

Hardware Efficient configuration:

repetitions: 2

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6. Execution Modes

The benchmark suite defines three execution environments.

Mode	Description
Ideal	Perfect statevector simulation

Shots	Shot-based sampling without noise
Noisy	Simulation with hardware noise model

Each benchmark run must specify the execution mode.

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7. Measurement Configuration

All benchmark runs must use the following measurement configuration.

Measurement Strategy

Pauli grouping

Shot Budget

8192 shots

This ensures consistent measurement overhead across experiments.

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8. Classical Optimizer

The benchmark suite defines a fixed classical optimizer.

Optimizer:

COBYLA

Maximum iterations:

200

Optimizer settings must not be modified in certified benchmark runs.

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9. Required Metrics

Each benchmark result must report the following metrics.

Metric	Description
vqe_energy	Estimated ground state energy
exact_energy	Reference exact energy
gap_ideal	Energy gap under ideal simulation
gap_shots	Energy gap under shot sampling
gap_noisy	Energy gap under noisy simulation
circuit_depth	Total circuit depth
two_qubit_gates	Number of two-qubit gates

measurement_groups	Number of grouped measurements

Benchmark entries missing required metrics cannot be classified as **validated or certified**.

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10. Benchmark Configuration Format

Benchmark configurations must include the following parameters.

molecule

basis

active_space

encoding

ansatz

execution_mode

optimizer

shots

measurement_strategy

Each configuration must generate a unique configuration hash.

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11. Benchmark Result Classification

Benchmark results are assigned trust levels.

Level	Description
Experimental	Development runs
Validated	Schema-compliant benchmark runs
Certified	Runs matching the official benchmark specification

Only certified runs may be used in official benchmark reports.

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12. Benchmark Suite Expansion

Future benchmark suite versions may include:

- additional molecular systems
- larger active spaces
- additional ansatz families
- real hardware execution

New benchmark definitions will be published under versioned specifications.

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13. Reproducibility Requirements

To reproduce certified benchmark results, the following must be provided:

- benchmark configuration file
- algorithm parameters
- measurement configuration
- optimizer settings
- execution environment

Reproducible benchmark execution must generate identical benchmark configuration hashes.

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14. Citation

If the QEncode Benchmark Suite is used in research or evaluation studies, please cite the QEncode whitepaper.